Intro to Machine Learning
Quiz 3 Practice Problems
Practice problems, not turned in — Solutions

Directions: These practice problems cover the material for Quiz 3: linear discriminant analysis in Part 1, quadratic discriminant analysis in Part 2, and basis functions and splines in Part 3. They are not turned in, but the quiz draws on them, so work through each one before looking at the solutions. Problems adapted from ISL cite the exercise; see the book for the original wording.

Part 1: Linear Discriminant Analysis

Notation follows ISL 4.4. Classes are $k = 1, \dots, K$, $\pi_k = \Pr(Y = k)$ is the prior, $f_k(x)$ is the density of X in class k, and $p_k(x) = \Pr(Y = k \mid X = x)$ is the posterior. When fitting LDA by hand, use the pooled variance with divisor $n - K$.

1. **The LDA discriminant.** (ISL 4.8, Exercise 2; ISL 4.4.1–4.4.2, ESL 4.3.)

   (a) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma^2)$, with the same σ^2 in every class. Show that the Bayes classifier assigns x to the class with the largest

   $$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k.$$

   (b) With $K = 3$ and $p = 4$, how many parameters does LDA estimate (priors, means and the shared covariance)? How many coefficients does multinomial logistic regression estimate? The LDA posterior has the same form as a multinomial logistic regression, so what is LDA doing with the extra parameters?

   Solution.

   (a) The Bayes classifier picks the k with the largest posterior, $p_k(x) = \pi_k f_k(x) / \sum_l \pi_l f_l(x)$. The denominator doesn't depend on k, so it picks the largest $\pi_k f_k(x)$, or equivalently the largest log:

   $$\log \pi_k + \log f_k(x) = \log \pi_k - \log\left(\sqrt{2\pi}\,\sigma\right) - \frac{(x - \mu_k)^2}{2\sigma^2}$$

   $$= \log \pi_k - \log\left(\sqrt{2\pi}\,\sigma\right) - \frac{x^2}{2\sigma^2} + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2}.$$

   The terms $-\log(\sqrt{2\pi}\,\sigma)$ and $-x^2/(2\sigma^2)$ are the same for every class (because σ is shared), so they don't change which k is largest. Dropping them leaves $\delta_k(x)$.

(b) LDA: $K-1=2$ priors, $Kp=12$ means and $p(p+1)/2=10$ covariance entries, so 24 parameters. Logistic regression: $(K-1)(p+1)=10$. Both end with a posterior that has 10 free coefficients (the same form), but LDA also models how X is distributed, which logistic regression leaves alone. If that model is right, the extra structure lets LDA use every observation to estimate the coefficients, so it has lower variance. If it is wrong, the coefficients are biased (the LDA vs. logistic lab, Sections 2 and 3).

2. **LDA with one predictor.** (ISL 4.8, Exercise 7; ISL 4.4.2.) We want to predict whether a stock will issue a dividend this year (yes or no) from X, last year's percentage profit. Among companies that issued a dividend, X averaged $\bar{x}=10$; among those that didn't, $\bar{x}=0$. The variance of X in both groups is $\hat{\sigma}^2=36$, and 80% of companies issued dividends. Assume X is normal in each group. Find the probability that a company with $X=4$ issues a dividend this year.

Solution. By Bayes' theorem, with the normal density's constant $1/(\sqrt{2\pi}\,\sigma)$ cancelling,

$$\hat{\Pr}(\text{dividend} \mid X=4)=\frac{0.8\,e^{-(4-10)^2/72}}{0.8\,e^{-(4-10)^2/72}+0.2\,e^{-(4-0)^2/72}}$$

$$=\frac{0.8\,e^{-0.5}}{0.8\,e^{-0.5}+0.2\,e^{-0.222}}=\frac{0.4852}{0.4852+0.1601}=0.752.$$

Even though 4 is closer to the "no" mean of 0 than to 10, the 4-to-1 prior for "yes" wins.

Comparing numerators only. To *classify* the company, you don't need the probability. The posterior for each class is $\pi_k f_k(4)$ divided by the same denominator, so the larger numerator wins:

$$\text{dividend: } 0.8\,e^{-(4-10)^2/72}=0.8\,e^{-0.5}=0.485,$$

$$\text{no dividend: } 0.2\,e^{-(4-0)^2/72}=0.2\,e^{-0.222}=0.160.$$

(Both still carry the common factor $1/(\sqrt{2\pi}\,\sigma)=1/(6\sqrt{2\pi})$, which doesn't change which is larger.) The dividend numerator is about three times the other, so LDA predicts a dividend. To turn the comparison into the probability, divide one numerator by their sum: $0.485/(0.485+0.160)=0.752$, as above.

3. **Fitting LDA by hand.** (ISL 4.4.2.) A training set has one predictor:

$$\text{class 1: } x=1,2,3; \qquad \text{class 2: } x=5,6,6,7.$$

(a) Compute $\hat{\pi}_k$, $\hat{\mu}_k$ and the pooled variance $\hat{\sigma}^2$.

(b) Find the LDA decision boundary. Which way did the unequal priors move it from the midpoint of the means?

Solution.

(a) $n=7$, $K=2$. $\hat{\pi}_1=3/7$, $\hat{\pi}_2=4/7$; $\hat{\mu}_1=2$, $\hat{\mu}_2=6$. The within-class sums of squares are $1+0+1=2$ and $1+0+0+1=2$, so $\hat{\sigma}^2=(2+2)/(7-2)=0.8$.

(b) From Problem 1(a), $\delta_1(x) = 2x/0.8 - 4/1.6 + \log(3/7)$ and $\delta_2(x) = 6x/0.8 - 36/1.6 + \log(4/7)$. Setting them equal, $5x = 20 + \log(3/4)$, so $x^* = 4 - 0.058 = 3.942$. Class 2 has the larger prior, so the boundary moves from the midpoint 4 toward class 1, the rarer class. With the boundary closer to class 1, class 2 is predicted over a wider range of x: every $x > 3.942$, not just $x > 4$.

Part 2: Quadratic Discriminant Analysis

QDA assumes $X \mid Y = k \sim N(\mu_k, \Sigma_k)$; naive Bayes assumes $f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$.

4. **QDA is quadratic.** (ISL 4.8, Exercise 3; ISL 4.4.3, ESL 4.3.) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$, with a different variance in each class. Find the discriminant $\delta_k(x)$ the Bayes classifier maximizes.

   Solution. As for LDA, the Bayes classifier maximizes $\log \pi_k + \log f_k(x)$. Now

   $$\log \pi_k + \log f_k(x) = \log \pi_k - \tfrac{1}{2}\log(2\pi) - \log \sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}.$$

   Only $-\frac{1}{2}\log(2\pi)$ is the same for every class, so

   $$\delta_k(x) = -\frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2} - \log \sigma_k + \log \pi_k.$$

   The $-x^2/(2\sigma_k^2)$ term depends on k because σ_k does, so unlike LDA it can't be dropped: $\delta_k(x)$ is quadratic in x.

5. **QDA by hand: same means, different spreads.** (ISL 4.4.3.) Two equally likely classes have $X \mid Y = 1 \sim N(0, 1)$ and $X \mid Y = 2 \sim N(0, 4)$.

   (a) Find the Bayes decision rule. Where is each class predicted?

   (b) Now let $\pi_1 = 0.8$. How does the rule change?

   Solution.

   (a) From Problem 4 with $\mu_1 = \mu_2 = 0$, $\sigma_1 = 1$, $\sigma_2 = 2$ and equal priors (which cancel), $\delta_1(x) = -x^2/2$ and $\delta_2(x) = -x^2/8 - \log 2$. Predict class 1 when

   $$-\frac{x^2}{2} > -\frac{x^2}{8} - \log 2 \iff \frac{3x^2}{8} < \log 2 \iff |x| < \sqrt{\tfrac{8\log 2}{3}} = 1.36.$$

   Class 1 is predicted on $(-1.36, 1.36)$ and class 2 on both sides of it. The boundary is two points, and class 2's region is two separate pieces.

   (b) The prior term $\log(\pi_1/\pi_2) = \log 4$ now favours class 1: predict class 1 when $3x^2/8 < \log 2 + \log 4 = \log 8$, that is $|x| < \sqrt{8\log 8/3} = 2.35$. Class 1's interval widens.

6. **LDA or QDA?** (ISL 4.8, Exercise 5; ISL 4.4.3, 4.5.) Answer each and explain.

(a) If the Bayes decision boundary is linear, do we expect LDA or QDA to do better on the training set? On the test set?

(b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to do better on the training set? On the test set?

(c) In general, as n increases, do we expect the test accuracy of QDA relative to LDA to improve, decline or stay the same?

(d) True or false: even if the Bayes boundary is linear, QDA will probably achieve a lower test error rate than LDA, because QDA is flexible enough to model a linear boundary.

(e) With $p = 10$ predictors and $K = 3$ classes, how many covariance parameters do LDA, QDA and Gaussian naive Bayes each estimate? How many observations does each class need for QDA's $\hat{\Sigma}_k$ to be invertible?

Solution.

(a) QDA on the training set: it is more flexible and can always match LDA's boundary, so it fits the training data at least as well. LDA on the test set: a linear boundary is right, so QDA's extra flexibility buys no reduction in bias, only extra variance.

(b) QDA on the training set (for the same reason as in (a)). On the test set, usually QDA, if n is large enough to estimate the Σ_k well: LDA's linear boundary is biased, and that bias doesn't shrink with n. With small n and large p, LDA can still win.

(c) Improve. QDA's disadvantage is variance, which shrinks as n grows; any advantage it has is lower bias, which doesn't.

(d) False. If the Bayes boundary is linear, LDA's model is correct, so QDA's extra parameters add variance and no reduction in bias. QDA *can* represent the linear boundary, but its estimate will not be exactly linear, and it pays for the extra parameters in test error.

(e) Each Σ has $p(p+1)/2 = 55$ distinct entries. LDA: 55 (one shared matrix). QDA: $3 \times 55 = 165$. Gaussian naive Bayes: $Kp = 30$ variances (only the diagonal of each Σ_k). All three also estimate $Kp = 30$ means. A sample covariance computed from n_k points has rank at most $n_k - 1$, so QDA needs $n_k \geq p + 1 = 11$ in *every* class.

Part 3: Basis Functions and Splines

Here $I(\cdot)$ is the indicator, equal to 1 when its condition holds and 0 otherwise.

7. **Degrees of freedom and variance.** (ISL 7.3, 7.4.4; ESL 7.3; the lecture's pointwise-variance and polynomials-versus-splines sections.) A model with basis functions $b_1, \ldots, b_J$ and an intercept is fitted by least squares to n points, so the basis matrix $\mathbf{B}$ is $n \times (J+1)$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ with $\mathbf{H} = \mathbf{B}(\mathbf{B}^T\mathbf{B})^{-1}\mathbf{B}^T$, and the degrees of freedom are $\mathrm{tr}(\mathbf{H}) = J + 1$. Treat the x_i as fixed, with $\mathrm{Var}(\varepsilon_i) = \sigma^2$.

   (a) Give J for a degree-7 polynomial, a cubic spline with 5 knots, and a natural cubic spline with 5 interior knots.

   (b) Using $\mathrm{Var}\left(\hat{f}(x_i)\right) = \sigma^2 h_{ii}$, show that the average variance of the fit over the training points is $\sigma^2(J+1)/n$. With $n = 100$ and $\sigma^2 = 1$, what is it for $J = 3$, 7 and 19? Does the answer depend on which basis is used?

(c) A cubic spline with 4 interior knots and a natural cubic spline with 6 interior knots both have $J = 7$. Compare their total variance over the training points and where along x each puts it. Which would you use if f wiggles a lot in the middle of the data but is close to linear near the ends?

Solution.

(a) Polynomial: $J = d = 7$. Cubic spline: $J = K + 3 = 8$ (x, x^2, x^3 and one truncated power function per knot). Natural cubic spline: $J = K + 1 = 6$ (its $K + 2 = 7$ basis functions, counting both boundary knots, include the constant). The cubic spline is the most flexible of the three.

(b) Averaging the variances over the training points,

$$\frac{1}{n}\sum_i \mathrm{Var}\left(\hat{f}(x_i)\right) = \frac{\sigma^2}{n}\sum_i h_{ii} = \frac{\sigma^2}{n}\,\mathrm{tr}(\mathbf{H}) = \frac{\sigma^2(J+1)}{n},$$

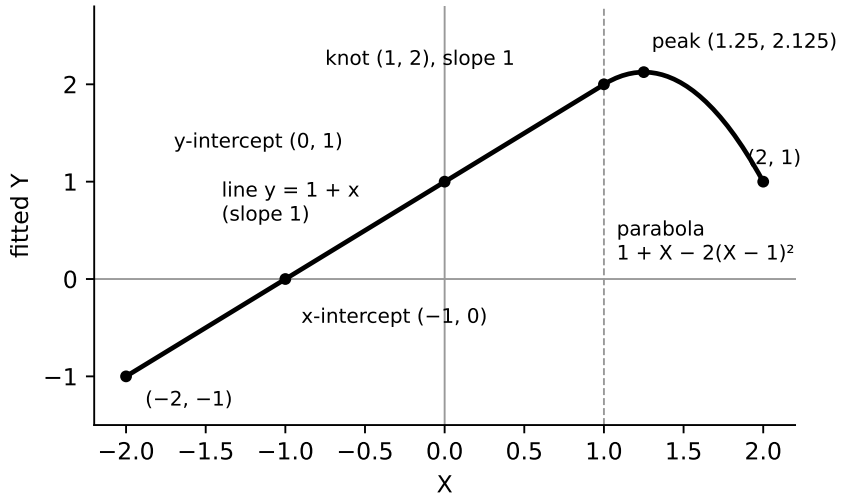
because $\mathrm{tr}(\mathbf{H}) = \mathrm{tr}\left((\mathbf{B}^T\mathbf{B})^{-1}\mathbf{B}^T\mathbf{B}\right) = \mathrm{tr}(I_{J+1}) = J + 1$. For $J = 3, 7, 19$: 0.04, 0.08, 0.20. It doesn't depend on the basis, only on J: the basis decides *where* along x the variance goes, not how much there is in total. More basis functions always means more variance, which is why adding flexibility only pays when it removes enough bias.

(c) Both have total variance $\sum_i \mathrm{Var}\left(\hat{f}(x_i)\right) = 8\sigma^2$, since that depends only on J. The cubic spline is a cubic beyond its last knots, so it puts more of its variance near the edges of the data; the natural spline is forced to be linear there, which cuts the edge variance, and spends the saved degrees of freedom on two more interior knots, so it has more variance in the middle. With f wiggly in the middle and nearly linear at the ends, the natural spline is the better choice: its extra interior knots lower the bias where f bends, and its linear tails cost little bias where f is nearly linear anyway.

8. **Sketching a fit from its basis functions, I.** (ISL 7.9, Exercise 3; ISL 7.3.) A curve is fitted with basis functions $b_1(X) = X$ and $b_2(X) = (X-1)^2 I(X \geq 1)$. The regression $Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon$ gives $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$ and $\hat{\beta}_2 = -2$. Sketch the fitted curve for $-2 \leq X \leq 2$, marking the intercepts, slopes and anything else of interest.

Solution. The fitted curve is $\hat{f}(X) = 1 + X - 2(X-1)^2 I(X \geq 1)$.

- For $X < 1$, $b_2(X) = 0$, so $\hat{f}(X) = 1 + X$: a line with slope 1, y-intercept $(0, 1)$ and x-intercept $(-1, 0)$. At $X = -2$ it is -1.
- For $X \geq 1$, $\hat{f}(X) = 1 + X - 2(X-1)^2$, a downward parabola. At $X = 1$ both pieces equal 2, and the parabola's slope $1 - 4(X-1)$ is 1, the same as the line's, so the curve has no jump and no kink at $X = 1$.
- The slope is 0 at $X = 1.25$, where the curve peaks at $\hat{f} = 2.125$. At $X = 2$, $\hat{f} = 1$.



9. **Sketching a fit from its basis functions, II.** (ISL 7.9, Exercise 4; ISL 7.2, 7.3.) A curve is fitted with basis functions

$$b_1(X) = I(0 \le X \le 2) - (X-1)I(1 \le X \le 2),$$

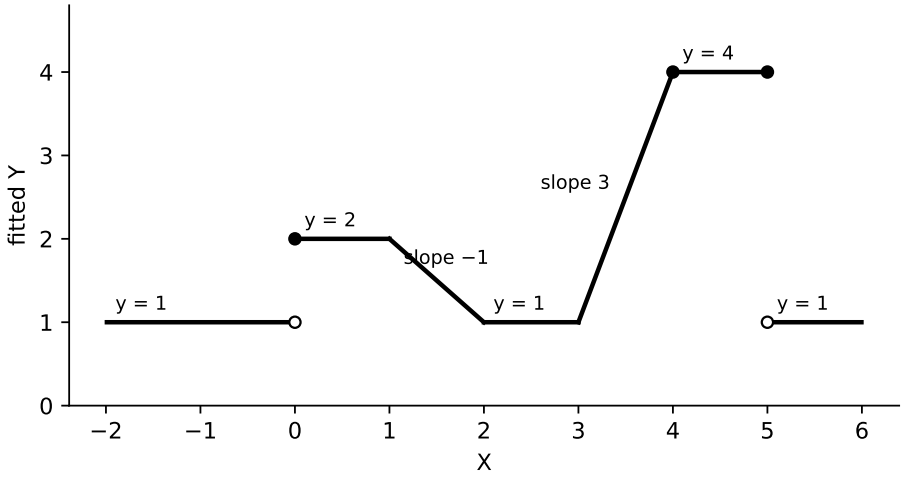
$$b_2(X) = (X-3)I(3 \le X \le 4) + I(4 < X \le 5).$$

The regression $Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon$ gives $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$ and $\hat{\beta}_2 = 3$. Sketch the fitted curve for $-2 \le X \le 6$, marking the intercepts, slopes and anything else of interest.

Solution. Work out each basis function piece by piece. $b_1(X) = 1$ on $[0, 1]$, $1-(X-1) = 2-X$ on $[1, 2]$, and 0 elsewhere. $b_2(X) = X - 3$ on $[3, 4]$, 1 on $(4, 5]$, and 0 elsewhere. Then $\hat{f}(X) = 1 + b_1(X) + 3b_2(X)$:

Interval	$\hat{f}(X)$	Shape
$-2 \le X < 0$	1	flat
$0 \le X \le 1$	2	flat
$1 \le X \le 2$	$3 - X$	slope -1, from 2 down to 1
$2 < X < 3$	1	flat
$3 \le X \le 4$	$1 + 3(X - 3)$	slope 3, from 1 up to 4
$4 < X \le 5$	4	flat
$5 < X \le 6$	1	flat

The y-intercept is $\hat{f}(0) = 2$. The curve jumps up from 1 to 2 at $X = 0$ and down from 4 to 1 just after $X = 5$, and is continuous everywhere else (at $X = 1$, 2, 3 and 4 the pieces meet).



10. **Exploring df with the basis expansions app.** (ISL 7.4.4–7.4.5; Problem 7.) Open the Basis Expansions app and press **Reset**, which gives the ESL wave $f(x) = \sin(12(x+0.2))/(x+0.2)$ with $\sigma = 1$ and $n = 100$. The bottom panel plots training MSE, LOOCV and test error against df for the selected basis; the app's df is $J + 1$ (the intercept counts). After Reset the sample is always the same, so your numbers should match the solutions.

    (a) With the **Polynomial** basis, move the degree from 1 to 15. Describe what happens to the training MSE, the test error and the $\pm 2\,\mathrm{SE}$ band. At what df is the test error smallest? What is the lowest the test error could ever be, and why?

    (b) Switch to the **Natural cubic spline** and move K from 0 to 15. At what df is the test error smallest? Compare the test error at the largest df with the polynomial's at its largest df. Which basis is hurt less by too much flexibility, and why?

    (c) For the natural spline, find two neighbouring values of K where the training MSE goes *up* as K increases. Why can that happen here, when the polynomial's training MSE never goes up?

    (d) Does LOOCV pick roughly the same df as the test error, for both bases? Which of the two curves could you compute on a real data set?

    (e) Keep the natural spline and set $n = 30$, then $n = 500$. How does the df with the smallest test error change? Explain using Problem 7. What happens to the polynomial at $n = 30$ once d reaches 8?

    Solution.

    (a) The training MSE goes down every time d increases (from 2.93 at $d = 1$ to 0.74 at $d = 15$). The test error drops sharply once the fit can follow the wave (from 2.56 at $d = 3$ to 1.24 at $d = 5$), is smallest at df $= 8$ ($d = 7$, test error 1.080), and then rises again, passing 2 by $d = 13$. The band is narrow in the middle and widens at the edges, and the widening at the edges gets much worse at high degree, which is where the extra test error comes from. The test error can never go below $\sigma^2 = 1$ (the grey line), the noise in a new Y that no fit can predict; the test error shown is $\sigma^2 + \int (\hat{f} - f)^2\, dx$.

(b) Smallest at df = 7 ($K = 5$, test error 1.083), about the same as the best polynomial. At the largest df the natural spline's test error is 1.19 (df = 17), against 1.58 for the polynomial at df = 16 (and above 2 at $d = 13$ and 14). The natural spline degrades much more slowly. A polynomial is global and adds flexibility by raising the degree, which inflates the variance at the edges; the spline adds knots, each of which changes the fit only locally, and its tails stay linear.

(c) For example, df = 12 to 13 ($K = 10$ to 11): 0.785 to 0.797. (Other pairs work too, such as df = 14 to 15.) When K changes, the knots are moved to new quantiles of x, so the model with $K + 1$ knots does not contain the model with K knots, and a non-nested bigger model can fit the training data worse. Polynomials of increasing degree are nested (every degree-d polynomial is a degree-$(d + 1)$ polynomial), so their RSS can only go down.

(d) Yes: LOOCV is smallest at df = 8 for the polynomial and df = 7 for the natural spline, the same as the test error. Only LOOCV can be computed on real data; the test error needs the true f, which the app knows because it made up the data. That is why we choose df by cross-validation.

(e) The best df goes from 6 at $n = 30$ to 7 at $n = 100$ and 9 at $n = 500$. The average variance $\sigma^2(J + 1)/n$ shrinks as n grows (Problem 7(b)), so each extra basis function costs less, while the bias it removes stays the same, and the minimum of the U moves to larger df. With $n = 30$, the polynomial's test error explodes from $d = 8$ on (about 24 at $d = 8$, over 600 at $d = 10$): with so few points near the edges, a high-degree polynomial swings wildly there.